

TABLE I
Electrical Characteristics of 1N914B Diode

Symbol	Parameter	Conditions	Min.	Max.	Unit
V_F	Forward voltage 1N914; 1N914A 1N914B 1N914B	$I_F = 10 \text{ mA}$	—	1	V
		$I_F = 5 \text{ mA}$	0.62	0.72	V
		$I_F = 100 \text{ mA}$	—	1	V
I_R	Reverse current	$V_R = 20 \text{ V}$	—	25	nA
		$V_R = 75 \text{ V}$	—	5	μA
		$V_R = 20 \text{ V}; T_J = 150^\circ\text{C}$	—	50	μA
C_d	Diode capacitance	$f = 1 \text{ MHz}; V_R = 0$	—	4	pF
t_r	Reverse recovery time	When switched from $I_F = 10 \text{ mA}$ to $I_R = 10 \text{ mA}; R_L = 100\Omega$; measured at $I_R = 1 \text{ mA}$	—	8	ns
		When switched from $I_F = 10 \text{ mA}$ to $I_R = 60 \text{ mA}; R_L = 100\Omega$; measured at $I_R = 1 \text{ mA}$	—	4	ns
V_{rr}	Forward recovery voltage	When switched from $I_F = 50 \text{ mA}; t_s = 20 \text{ ns}$	—	2.5	V

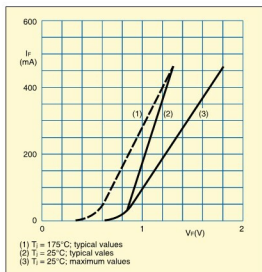


Fig. 5: Forward current vs forward voltage of 1N914B diode

ing that is to be considered. Repetitive peak inverse voltage rating of 1N4007 is 1000 volts.

For conventional single-diode half-wave rectifier (Fig. 1), two-diode full-wave rectifier (Fig. 2) and bridge rectifier configurations (Fig. 3), the chosen diodes should have a repetitive peak inverse voltage rating equal to at least the peak of the applied AC input, twice the peak value of the applied AC input and the peak value of the AC input, respectively. The diode to be used in the cascade type of diode-capacitor

multiplier chain (Fig. 4) should have PIV rating greater than the peak value of the AC signal applied at the input of the multiplier chain.

As an illustration, consider diode 1N4001 with peak repetitive inverse voltage of 50V. When this diode is used for a conventional half-wave or bridge rectifier, it can handle an AC signal with maximum peak value of 50V. As a two-diode full-wave rectifier it can handle an AC signal with maximum peak value of 25V only. For AC signals having higher peak values, other diodes like 1N4002 through

1N4007 can be used depending upon the exact value of the peak voltage of the AC signal.

Maximum average rectified current ($I_{F(av)}$). It is the maximum average forward rectified output current that can

be allowed to pass through the diode without exceeding its power dissipation limits. The average current rating is lower than the continuous or peak repetitive forward current.

For a half-wave rectified signal:

$$I_{F(av)} = I_{peak} / \pi$$

For a full-wave rectified signal:

$$I_{F(av)} = 2I_{peak} / \pi$$

where I_{peak} is the peak repetitive forward current. It is useful in selecting diodes for rectification applications.

The maximum average rectified current rating of the selected diode should be greater than the average current flowing through it in a particular application. The average rectified current rating of 1N4001-1N4007 series general-purpose rectifier diodes is 1A for 1cm lead length at an ambient temperature of 25°C .

Peak repetitive forward current. It is the maximum instantaneous value of the repetitive forward current.

Peak forward surge current. During turn-on, malfunction, switching, etc, high values of current may flow through the diode for brief time intervals. Surge current ratings define the maximum value and time duration of such surges in the current level. For instance, a surge rating of 10A for 10 ms implies that the diode can handle a maximum of 10A of forward current for time duration not exceeding 10 ms. The surge current rating is significantly higher than the maximum rectified current rating. This specification assumes more importance for switching diodes like 1N914 which have a peak forward surge current rating of 1A for a pulse width of 1 second and 4A for a pulse width of 1 μs .

Maximum junction temperature (T_J). It is the maximum allowable junction temperature. It is significant in

PIV, average rectified forward current and I^2t ratings are the parameters of interest in the case of rectifiers for 50/60Hz AC mains. The same is not true for rectifiers needed for switched-mode power supplies where reverse recovery time is of prime concern.